不在范围内。未声称具备专家金标准可比性标签、论文级定位器提取、生成推荐评估或实验室复现。未来工作必须在留存的智能体迹上校准检测器画像，按 DOI 和设备族进行分组划分，加入专家裁定，测试去中心化执行，并至少测试第二个适配器。

科学责任与更广泛影响。协议安全性以适配器可靠性为条件。不完整的领域规则可能接纳错误证据，而过度的弃权可能不成比例地压制报告不足的子领域。因此，EviCoord 记录拒绝原因，暴露未解决的不确定性，将高影响任务上未解决的严重冲突升级给人类科学家，并在低影响情况下弃权。它不能替代复现、安全评估、作者责任或机构审查。数据发布必须保留来源许可证，并避免分发受版权保护的全文。

人工智能使用披露。OpenAI Codex 助手用于整合项目笔记、辅助分析与可视化代码、检索候选文献及编辑稿件。所有数值声明均由已公开的可复现脚本生成；未虚构任何模型或专家实验。人类作者仍对研究设计、来源验证、代码、结果、解释及最终提交负责。

8 结论

EviCoord 将证据状态协调确立为多智能体 AI4S 的通用基础。类型化证据信封、可复用的核心-适配器分离、来源与可比性门控、冲突隔离、验证器多样性以及有界终止动作，将科学证据转化为可追踪、可验证且可治理的协调对象。由此产生的架构为跨领域部署提供了清晰路径：协调核心保持稳定，而专家知识通过明确的、领域特定的适配器进入。在基于真实记录的钙钛矿实例中，受控故障实验表明，自适应证据状态路由显著减少了不安全输出，同时保持了有用的覆盖范围并控制了验证成本。综合形式化保证、实现审计与实证结果，EviCoord 定位为一种实用且可扩展的协调范式，用于跨学科扩展可靠的多智能体科学发现。

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